

STA304 Practice Final Examination #2

Duration: 3 Hours Total Marks: 50

Part A: Long Answer

Question 1 – Survey Design and Selection Bias

A municipality wishes to estimate the average monthly spending on public transit among residents. The municipality is considering the following designs:

1. Surveying individuals exiting a downtown subway station during rush hour.
 2. Sending invitations to a simple random sample of residents from a municipal registry.
 3. Posting a voluntary survey link on social media.
- a) Define the target population.

b) Identify one potential source of selection bias for Design 1.

c) Identify one potential source of selection bias for Design 3.

d) Which design would you recommend? Justify your choice.

Question 2 – Stratified Sampling

A university wishes to estimate average weekly study hours.

Faculty	N_h	$\bar{y}_h$	s_h
Arts	1200	11	4
Science	800	15	8
Engineering	500	18	10

- a) Compute the stratified estimate of the population mean.
- b) Explain why proportional allocation may not be optimal in this setting.
- c) Determine the Neyman allocation proportions.
- d) Which faculty contributes most to the sampling variance? Explain.

Question 3 – Cluster Sampling

A school board selects 10 schools from a population of 200 schools.

Within each selected school, 20 students are sampled.

The sample means for the selected schools are:

62, 58, 65, 71, 68, 59, 63, 60, 66, 64

- a) Identify the PSUs and SSUs.
- b) Estimate the overall population mean.
- c) Explain why observations within the same cluster may be correlated.
- d) Describe the effect of a large ICC on variance estimation.

Question 4 – Ratio and Regression Estimation

A simple random sample of farms is selected.

$$\bar{y}_S = 180$$

$$\bar{x}_S = 75$$

$$\bar{x}_U = 90$$

a) Compute the ratio estimate.

b) Is the ratio estimate larger or smaller than $\bar{y}_S$? Explain.

c) Under what conditions is ratio estimation preferable to the ordinary sample mean?

d) Suppose

$$y_i = 50 + 2x_i + \varepsilon_i.$$

Would you generally prefer a ratio estimator or regression estimator? Explain.

Question 5 – Unequal Probability Sampling

A population contains four units.

Unit	y_i	π_i
1	10	0.20
2	20	0.30
3	30	0.40
4	40	0.60

Suppose units 2 and 4 are selected.

- a) Compute the total estimate.
- b) Explain why unequal probability sampling may improve precision.
- c) What characteristics make a variable useful as a size measure?

Question 6 – Missing Data and Nonresponse

A health survey records age group for all sampled individuals but receives incomplete responses to a question about weekly exercise.

Age Group	Response Rate
18-34	90%
35-54	70%
55+	45%

- a) Does this pattern provide evidence against MCAR? Explain.
- b) Can these data distinguish MAR from NMAR? Explain.
- c) Describe how post-stratification could be used to adjust for nonresponse.
- d) Describe the two-stage approach to nonresponse adjustment discussed in class.

- e) Name one additional variable that would likely be useful in a response propensity model and explain why.

Part B: Multiple Choice

1. Which statement about domain estimation is true?
 - (A) Domains must be specified before sampling.
 - (B) Domain sample sizes are fixed by design.
 - (C) Domains are defined after sampling and used for analysis.
 - (D) Domains require proportional allocation.
 - (E) Domains eliminate bias.
2. Which situation most clearly demonstrates undercoverage?
 - (A) A survey has a 60% response rate.
 - (B) A survey excludes individuals without internet access.
 - (C) Respondents skip one question.
 - (D) Interviewers influence responses.
 - (E) Volunteers choose whether to participate.
3. Which allocation method assigns larger samples to strata with larger variability?
 - (A) Equal allocation
 - (B) Proportional allocation
 - (C) Neyman allocation
 - (D) Domain allocation
 - (E) Post-stratification
4. A large positive ICC implies:
 - (A) Clusters are internally heterogeneous.
 - (B) Cluster sampling becomes more efficient than SRS.
 - (C) Individuals within clusters tend to be similar.
 - (D) The FPC increases.
 - (E) Selection bias is present.
5. Which scenario is most consistent with NMAR?
 - (A) Missingness depends only on age.
 - (B) Missingness depends on observed sex.
 - (C) Missingness depends directly on the unobserved response value.
 - (D) Missingness is unrelated to any variables.
 - (E) Missingness depends only on sampling weights.

6. Which question is a leading question?
- (A) How many hours do you study?
 - (B) What is your age?
 - (C) Do you support the university's sensible plan to improve student life?
 - (D) What is your major?
 - (E) How often do you use public transit?
7. Under SRS, when is the FPC most important?
- (A) Small sampling fraction
 - (B) Large sampling fraction
 - (C) Infinite population
 - (D) Sampling with replacement
 - (E) Cluster sampling
8. Ratio estimation is generally most effective when:
- (A) x and y are negatively related.
 - (B) x and y are strongly correlated.
 - (C) Population size is unknown.
 - (D) Sampling fractions are small.
 - (E) Stratification is impossible.
9. Which estimator is generally preferred when the relationship contains a substantial non-zero intercept?
- (A) Ratio estimator
 - (B) Regression estimator
 - (C) Domain estimator
 - (D) Cluster estimator
 - (E) Post-stratified estimator
10. Which statement about observational studies is correct?
- (A) They always establish causation.
 - (B) They never provide useful evidence.
 - (C) Confounding variables may explain observed associations.
 - (D) Randomization is guaranteed.
 - (E) Selection bias cannot occur.

11. In PPS sampling, units with larger size measures:

- (A) Always have equal selection probabilities.
- (B) Tend to have larger selection probabilities.
- (C) Are excluded from sampling.
- (D) Have smaller weights.
- (E) Cannot be selected more than once.

12. Self-selection bias is most likely when:

- (A) Respondents are randomly selected.
- (B) A probability sample is used.
- (C) Participation is entirely voluntary.
- (D) A census is conducted.
- (E) Post-stratification is used.

Answer Key

MCQ: 1(C), 2(B), 3(C), 4(C), 5(C), 6(C), 7(B), 8(B), 9(B), 10(C), 11(B), 12(C)